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Cyber-Security Awareness Report

Mini Project 4 · Team Report

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1.1 Introduction

Section lead: Henry Peer reviewed by: Wayne

1.2 Purpose and Importance

Cybersecurity is becoming ever more important in our modern world. As technology improves and becomes more and more a part of our daily lives, the risks we take as we use these devices also increase accordingly. These risks can cause things like identity theft, scams, and malicious software; This brief will be on the different methods you can take to secure yourself, the effectiveness of said methods as well as the phycology methods that bad agents may use to try and get information out of you.

1.3 Scope and Intended audience

This report covers 5 different sectors including the threats, prevention and reaction methods that are utilised when discussing cybersecurity. The report is intended for people with a minimal understanding of cybersecurity awareness so they can better recognise threats and prevention methods to keep themselves safer. Its also intended for organisations to get a better general understanding on how to keep their systems safe from internal or external threats.

2.1 Passwords

Section lead: Haneef Peer reviewed by: Pandeli

2.2 What is a Password?

A secret combination of alphanumerical characters and symbols used to verify identity and grant access to computer systems, online accounts and secured computer resources.

The idea is to protect personal information, data (such as personal identity, medical records, payment details etc...) and accounts from unauthorised access.

Secure practices when creating, storing, and using passwords. For example, the use of long, unique passwords for every new or existing account, and enabling MFA or 2FA to reduce the risk of unauthorised access.

Additionally, here are a list of elements it includes:

- Ensure the password is unique across all accounts
- Refrain from using predictable patterns (such as, names, dates etc)

- Use of complex, strong and long passwords (12-16+ characters)
- Enabling Multi-Factor Authentication (MFA) as mentioned above
- Update passwords when suspected of compromise
- Check breach databases for recognised passwords recorded in breaches Why Password Hygiene matters?

Poor password hygiene, weak passwords or reuse, unsafe password storage are currently one of the largest weaknesses in modern day cybersecurity. Attackers often make use of leaked credentials in stuffing attacks. Attackers will rely on the fact that people reuse the same password across various of different sites, platforms and services. Which is what Password Hygiene is built to prevent.

2.3 Benefits of Password Hygiene

2.3.1 Unique passwords

By using separate passwords across all platforms, sites and services you essentially eliminate the one advantage cyber attackers hope to have when executing their exploits (such as password stuffing). It greatly lowers the risk of unauthorised access.

Strong and long passwords

Using strong and long passwords makes it more difficult for attackers to guess or crack them. Short passwords and common words are easier to compromise, especially when attackers use automated tools to test thousands or millions of possible combinations.

A strong password should ideally be at least 12–16 characters long. Using a mixture of letters, numbers and symbols can help, but length and unpredictability are very important. Another good option is using a passphrase made up of several unrelated words, to make it unique and difficult to predict.

2.3.2 Multi-Factor Authentication (MFA)

Multi Factor Authentication adds another layer of security to an account. Instead of relying only on a password, the user must provide another form of verification, such as an authentication app, a security key or a code sent to their device.

This means that even if an attacker manages to obtain a password, they may still be unable to access the account without the additional verification method. Protection against data breaches

Websites and online services can experience data breaches, which may expose users personal information and login credentials. If a password has been exposed in a breach, attackers may attempt to use it to access other accounts belonging to the same person.

Good password hygiene helps reduce this risk by ensuring that passwords are unique and that compromised credentials are changed quickly. Users should also check whether their email addresses or passwords have appeared in known data breaches. Improved privacy and data protection

Passwords help protect sensitive information, including personal details, financial records, private messages and work-related documents. Maintaining good password hygiene reduces the chances of this information being accessed by unauthorised individuals.

This is especially important in workplaces where employees may have access to confidential or sensitive company information, customer records or internal computer systems.

2.4 Common Password Security Risks

2.4.1 Password reuse

Password reuse occurs when a person uses the same password for multiple accounts. This is risky because a single compromised account could allow an attacker to attempt access to several other services. (Like by using password stuffing)

For example, if someone uses the same password for their email, social media and online banking, a breach on one website could put the other accounts at risk. Using a different password for every account helps prevent this.

2.4.2 Weak and predictable passwords

Passwords such as "123456", "password" or a person's name are easy to guess. Attackers may also use personal information, such as birthdays, pet names or favourite teams, to make educated guesses.

Users should avoid information that can easily be discovered through social media or other public sources. Instead, passwords should be unpredictable and difficult for other people to associate with the account owner.

2.4.3 Phishing attacks

Phishing is a technique in which attackers pretend to be legitimate organisations or individuals to trick users into revealing sensitive information. They may send emails or text messages containing links to fake login pages.

If a user enters their password on one of these pages, the attacker may capture the credentials and use them to access the real account.

To reduce this risk, users should check website addresses carefully, avoid suspicious links and never enter passwords on websites they cannot verify as legitimate.

2.4.4 Unsafe password storage

Storing passwords in plain text documents, unprotected notes or easily accessible files can expose accounts if a device is lost, stolen or compromised.

A password manager is a useful alternative. It securely stores passwords and can generate strong, unique credentials for different accounts. Users should protect their password manager with a strong master password and enable MFA where available.
Best Practices for Password Hygiene

2.4.5 Use a password manager

A password manager helps users create, store and manage unique passwords without needing to memorise every single one. It can also help identify weak or reused passwords and generate stronger alternatives.

Users should choose a reputable password manager and protect access to it carefully.

2.4.6 Change passwords when necessary

Passwords do not always need to be changed at regular intervals if they remain strong and secure. However, they should be changed immediately if there is evidence that they have been compromised, exposed in a breach or accessed by an unauthorised person.

Users should also change default passwords on devices and systems, as these may be publicly known or easy to guess.

2.4.7 Keep recovery options secure

Account recovery methods, such as backup email addresses, phone numbers and recovery codes, should also be protected. If an attacker gains access to a recovery account, they may be able to reset passwords and take control of other accounts.

Recovery codes should be stored securely and should not be shared with other people.

2.4.8 Check for compromised passwords

Users can check whether their email addresses or passwords have appeared in known data breaches through reputable breach checking services. If a password has been exposed, it should be replaced with a new, unique one.

Users should never enter their current password into an unfamiliar website simply because it claims to check password security.

2.5 Conclusion

Password hygiene is an important part of cybersecurity because passwords are often the first line of defence against unauthorised access. Weak passwords, password reuse

and unsafe storage can expose personal information and allow attackers to compromise multiple accounts.

By creating long and unique passwords, using a password manager, enabling Multifactor Authentication and responding quickly to suspected compromises, users can significantly improve their account security.

Ultimately, good password hygiene is not just about creating a strong password. It is about developing safe habits when creating, storing and using passwords to protect personal information, workplace systems and digital resources.

7 Password Hygiene Best Practices to Follow | Dashlane What is password stuffing?

What is credential stuffing? Meaning, examples, and prevention

I also utilised the AI overview for some of my research mainly for much simpler topics such as the risks of not using password hygiene.

3.1 2FA

Section lead: Brain Peer reviewed by: Haneef

Two factor authentication (2FA) and multi factor authentication (MFA) are security procedures and mechanisms that are designed to reduce the risk of unauthorised account access. Traditional password only authentication relies on a single piece of information that can be stolen, guessed or obtained through other ways like phishing and social engineering attacks. MFA adds an additional layer of security and authentication factor. Meaning that just possessing the password of an account shouldn't be enough to access an account and the information on it. The National Institute of Standards and Technology (NIST) defines three traditional authentication factor categories, something you know, something you have, and something you are (NIST, 2017).

3.2 Authentication factors and common methods

The first category, “something you know”, refers to information that is known by the user, such as a PIN or password. Passwords are widely used but can be vulnerable to phishing attacks, password reuse, credential stuffing and data breaches.

The second category, “something you have”, refers to an item or device controlled by the user, such as a mobile phone, authenticator app, or a physical security key.

The third category, “something you are”, refers to biometric characteristics, such as fingerprints, facial recognition technology and retinal scans. NIST explains that MFA involves using more than one of these different factor categories (NIST, 2017).

Common MFA methods usually include SMS codes, authenticator applications, and hardware security keys.

SMS authentication sends a one-time code to the user's registered mobile phone number, that is valid for a limited amount of time before being deleted and unusable. Although convenient and widely supported. SMS still does have security weaknesses. Attackers can potentially take control of a victim's mobile phone number through SIM swapping and some weaknesses in telecommunication infrastructure can also expose SMS authentication codes and be intercepted and used. NIST therefore classifies the use of SMS and voice calls through the public telephone network as a restricted authentication method because attackers have demonstrated the ability to obtain reassigned telephone numbers and intercept authentication messages (NIST, 2025).

Authenticator applications, such as Microsoft authenticator or Google authenticator. Generate one-time passwords or provide authentication prompts. These methods are generally stronger than traditional SMS one-time codes because they do not depend on the telephone network. However, one-time codes can still be stolen through phishing. The UK National Cyber Security Centre (NCSC) therefore places app-based authentication method below phishing resistant FIDO2 authentication in its recommended hierarchy (NCSC, 2024a).

Hardware security keys provide another form of “something you have”. Technologies such as FIDO2 use a public key cryptography and are designed to resist phishing because the authentication credential is cryptographically associated with the legitimate website. CISA describes security keys as providing strong protection against phishing, while NSCS identifies FIDO2 credentials as providing phishing and theft resistance (CISA, 2025) (NCSC, 2024a)

3.3 How effective is MFA at preventing account takeover?

Evidence from a Microsoft study indicates that MFA can substantially reduce the risk of account compromise, although its effectiveness depends on the methods of authentication used. A Microsoft study analysed suspicious activity involving Microsoft Azure Active Directory accounts between April and September 2022, the researchers estimated that MFA reduced the risk of account compromise by 99.22% across the

overall population and by 98.56% for accounts where credentials had been leaked and compromised completely. The study also found that authenticator applications performed better than SMS authentication (Lucas A. Meyer, 2023)

This study provides useful evidence because it examined real account activity rather than relying solely on theoretical security models. However, the results should not be interpreted as meaning that every MFA implementation prevents 99.22% of attacks. Rather the research examined Microsoft's commercial account environment and a specific period of suspicious activity, so the results may not directly correlate and represent every organisation, individual or type of MFA.

Earlier research conducted by Google with researchers from New York University and University of California, San Diego also demonstrated the effectiveness of additional authentication challenges. In their study, SMS based challenges blocked 100% of automated bot attacks, 96% of bulk phishing attacks and 76% of targeted attacks in the investigated dataset. Device based prompts performed better against targeted attacks, blocking 90% of them (Thomas K., 2019)

These findings demonstrate an important distinction; MFA is not equally effective regardless of the method used. Stronger phishing resistant methods can provide greater protection than SMS or methods that require users to enter codes into websites.

3.4 Why do security organisations still recommend SMS MFA?

Despite its weaknesses, security organisations still continue to recommend enabling MFA, including SMS based MFA when stronger alternatives are unavailable. This is because SMS MFA still provides that additional barrier compared with the classic password only authentication method. The UK NCSC explicitly states that SMS is not the most secure type of 2 step verification but still provides a significant advantage over not using a second factor at all (NCSC, 2025)

There are also practical reasons for continuing to support SMS. Almost every mobile phone can receive text messages, meaning that users do not necessarily need to install a secondary application or purchase specialised hardware security keys, which could make things a lot more complicated. This makes SMS particularly useful for legacy systems and users who cannot access more advanced authentication methods and

solutions. The NCSC therefore recommends message-based authentication primarily where stronger methods are not possible (NCSC, Why MFA matters, 2024b)

However, organisations should recognise the limitations of SMS authentication. CISA recommends stronger phishing resistant MFA methods, such as hardware-based FIDO authentication, and describes SMS as a weaker option (CISA, 2025). Therefore, SMS can be considered a practical minimum layer of security and protection rather than the strongest available MFA solution.

Overall, MFA significantly strengthens account security by requiring evidence from multiple authentication factors. Research from Microsoft and Google demonstrates that MFA can substantially reduce account compromise, but the level of protection varies according to the authentication method. SMS provides useful and the most simple and easy to deploy protection against many automated and credential-based attacks but remains vulnerable to more advanced techniques such as SIM swapping and phishing. Authenticator applications generally provide stronger protection, while FIDO2 security keys provide phishing resistant authentication. Consequently, organisations should implement MFA whenever possible while selecting authentication methods according to the security requirements, usability needs and risks associated with their systems.

4.1 Phishing

Section lead: Henry Peer reviewed by: Brian

4.1.1 Understanding phishing

Phishing remains a primary threat vector within corporate networks, functioning as an entry point for widespread infrastructure compromise. Fundamentally, phishing is a form of social engineering where an adversary masquerades as a trustworthy entity to manipulate targets into surrendering credentials, disclosing confidential information, or initiating unauthorized financial transactions. While traditional phishing campaigns are distributed indiscriminately to broad demographics, attackers increasingly deploy specialized variants designed to exploit specific structural channels.

4.1.2 Defining these variants

from requires precision. Spear phishing replaces mass distribution with targeted reconnaissance, using bespoke details gathered from open-source intelligence to victimise specific individuals. When spear phishing targets corporate executives, it is classified as whaling. Attackers also diversify beyond email protocols. Smishing exploits Short Message Service text channels, relying on the high trust users place in

mobile devices. Conversely, vishing utilizes voice over IP telephony or artificial intelligence voice-cloning to simulate urgent verbal commands. The most financially destructive permutation is Business Email Compromise (NCSC, Ncsc Buisness Payment Fraud, 2024). BEC attacks often omit malicious files entirely, relying instead on compromised or spoofed e-mail accounts from those higher in the chain of command to hijack the existing corporate workstations and divert corporate funds wherever the attacker wants.

4.2 The way phishing works

The operational efficiency of these attacks relies heavily on exploiting established human phycological biases. Some social engineering literature indicates that security measures are frequently bypassed by manipulating a target's psychological state. Attackers heavily exploit urgency by enforcing artificial, condensed deadlines. This structural constraint forces individuals into rapid, heuristic decision-making, which severely degrades their analytical faculties (S. Grazioli, 2000).

This state is routinely amplified by projecting an aura of authority. As demonstrated in foundational psychological paradigms, individuals are socially conditioned to comply with instructions issued by perceived superiors, this is notable demonstrated in the famous Milgram experiment (Milgram, 1963). In a corporate setting, a message mimicking an executive or some form of legal entity instinctively dampens a subject's inclination to verify the request. Furthermore, attackers exploit fear and loss aversion by threatening professional reprisal or system suspension. Alternatively, they may exploit helpfulness or curiosity. These intense emotional triggers effectively blind targets to contextual anomalies.

4.3.1 Identifying phishing

To identify these fraudulent communications, international bodies like the UK National Cyber Security Centre (NCSC, Annual Review 2025, 2025) and the US Cybersecurity and Infrastructure Security Agency (CISA, SECURING HIGH VALUE ASSETS, 2018) provide formal guidance detailing core indicators of deception. A primary indicator is a discrepancy within sender identities. Attackers routinely employ 'typo squatting' to mimic corporate domains or forge display names to mask external email addresses. Examining the absolute header data rather than the friendly display name reveals these variances.

Furthermore, the nature of the request itself serves as an indicator. Phishing attempts often look for immediate credential validation e.g.: changes to banking details, or the execution of unusual internal processes usually bypassing security methods. While early guidance focused on poor spelling or grammatical errors, modern adversaries utilize generative artificial intelligence to produce linguistically polished messages that

mimic the usual formal emails sent. Consequently, contemporary guidance highlights structural anomalies. These include generic greetings in place of personalized communication, mismatched destinations when hovering over hyperlink text, and requests that actively circumvent standard internal approval pathways. This can often times be the phishing messages greatest weakness as anyone with the know-how to check for such details or any company with a security setup telling the user if the message originated from inside or outside the company can show the phishing messages true intent.

4.3.2 Preventing phishing

When a phishing interaction is suspected it's important that the organisation or individual does not click on any link and instead report it to the NCSC by emailing report@phishing.gov.uk and let their takedown service investigate, hunt down and ultimately blacklist the suspicious email address if its found to be a phishing scam.

It's also important to block the email address so it doesn't pose any further security risk (e.g. if one department is sent an email then doesn't block it, then another department that is unaware of the risk falls for the scam it could pose a continuous security risk.

4.4 Recommended Response

When a phishing incident occurs within an organization, they must trigger an immediate incident response framework spanning individual and organizational dimensions.

An individual who identifies a suspected message before interacting with it must completely isolate the message. This means they must not reply to the sender open any links, download attachments. If an individual has already clicked a link or submitted data, they must immediately disconnect their endpoint from local wireless or wired networks to stop any potential malware from spreading laterally. The incident must then be reported through designated internal channels, such as a secure IT service desk telephone or a dedicated reporting button.

Organizationally, the security operations team must isolate the affected device from the broader network infrastructure. They must perform a comprehensive mail system sweep to purge identical messages from all corporate inboxes. If credentials were stolen, administrators must terminate active user sessions and enforce an enterprise-wide password reset, it's also a good idea to temporarily remove user group access to any sensitive data or systems from anyone but important IT responding staff. If a BEC attack has initiated a fraudulent wire transfer, the organization must instantly notify internal legal counsel and contact the banking institution to initiate a transaction freeze. Finally, a post-incident review must be completed to update email filtering rules and improve security awareness training modules based on the attack methodology.

5. Devices

Section lead: Wayne Peer reviewed by:

5.1 Safe use of devices

Use of devices is an important sector of being safe online because phones, computers, tablets will have personal information, passwords, photos, sensitive information and financial details. If you lose a device or its stollen, infected with malware or connected to an unsecure device the information can then be damaged or accessed. This also involves keeping software updated, protecting devices with screen locks and encryption using software and being careful when downloading apps. Reports from the national cyber security centre state that having secure devices is an important part of protecting users and organisations from online threats. Use of safe devices essentially involves to be keeping access to devices protected using security software and encryption, making sure software is kept up to date also in general being careful your USB devices or public charging points. The National Institute of Standards and Technology explains that devices can face threats such as theft, malware and unauthorised access to stored information (Scarfone, 2007).

5.2 Keeping Your Software Updated

A software vulnerability is a weakness or problem in software that can potentially be used by an attacker, and a patch is updating software which is deigned to fix the vulnerability or other problem, exploiting can be a method or code that takes advantage of the vulnerabilities. It is very important to keep your software updated because cyberattacks can use known vulnerabilities to gain access to people's data or devices. People should enable automatic updates when you can an install security patches to reduce risk.

The NCSC explained that software updates help fix known security flaws and reduce the chances of attackers compromising devices. (NCSC, Managing-Deployed-Devices, 2021a).

Another important point is making sure devices are supported by their makers, older devices may potentially stop receiving security updates which means newfound vulnerabilities might not be fixed. This can leave users exposed to cyberattacks even if they regularly check for updates. The NCSC (NCSC, Managing-Deployed-Devices, 2021a) recommends replacing unsupported devices and software where it can replace these risks.

5.3 Encryption, Security Software, Screen locks

Security software like antivirus software helps to protect devices by blocking and being able to identify dangerous programs therefore antivirus software needs to be updated on the regular so it can be able to spot new pieces of malware. The NSCS recommend using antivirus software and other security software as part of a larger approach to device protection (NCSC, Managing-Deployed-Devices, 2021b).

For example, an Employee may leave their laptop unattended when taking a break, without a screen lock another person may access important documents or personal information, using a strong password or pin and locking the screen when leaving the device helps reduce the risk of unauthorised access (NCSC, Managing-Deployed-Devices, 2021e).

5.4 Risk of Installing data from Invalid Sources

This shows users not to straight up trust every app or website, downloading stuff from trusted developers or official stores, checking the permissions of the applications and making sure software is updated can help decrease the risks. The NCSC recommends controlling which third party applications are installed on devices to reduce potential threats. (NCSC, Managing-Deployed-Devices, 2021d)

5.5 Unknown USB Devices and Public Charging Points

Public charging port points can have security risks because some USB can transfer data and power, if a charging station has been modified to have malicious intention and can be potentially used to manipulate a connected device. Research by Spolaor at ACM (Spolaor, 2016) showed that USB charging connections could be exploited to take information from devices. To decrease the risk, you could instead use your own charging plug or carry a power bank.

Safe use of devices is important because it helps to protect your personal information and decreases the risk of cyberattacks, keeping software updated lessens attackers from manipulating vulnerabilities, while screen locks and encryption protect information if a device is stolen or lost. Security software helps to provide more protection against malware and downloading apps from valid sources decreases unnecessary risks. People using should be careful connecting unknown USB devices or using public charging points, by successfully following these precautions organisation and individuals improve device security and decrease the chances of information being infiltrated.

6. Public Wi-fi

Section lead: Pandeli Peer reviewed by: Henry

6.1 Safe use of public WI-FI

Public Wi-Fi is available in many places such as universities, libraries, cafés, airports, hotels and shopping centres. It provides a convenient way for people to access the internet when they are away from home. However, public Wi-Fi can also create security and privacy risks if it is not used carefully. Users should understand these risks and take appropriate precautions to protect their personal information, accounts and devices. (FTC, Are public wifi networks safe what you need to know, 2023)

One of the main risks of public Wi-Fi is that not every network is trustworthy. Some networks may be poorly secured, while others could be deliberately created by attackers to trick people into connecting. For example, someone could create a Wi-Fi network with a name similar to a legitimate café or university network. A user may connect without checking whether it is the official network. Once connected, their internet activity could potentially be exposed to an attacker.

Another risk is the theft of personal information. When using an unsecured network, sensitive information such as usernames, passwords or other data may be at greater risk, particularly when using websites that do not properly protect communications. This is why users should avoid entering highly sensitive information, such as online banking details, while connected to an unfamiliar public network. If sensitive tasks are necessary, it is safer to use a trusted mobile connection or another secure network. (FTC, FTC Provides Tips for Using VPN Apps, 2018)

6.2 VPN protection and HTTPS

Users should also make sure that websites use HTTPS. HTTPS encrypts the communication between a user's browser and the website, making it more difficult for other people on the network to read the information being exchanged. Most modern browsers display a padlock or similar security indicator when a connection is protected by HTTPS. However, users should still be careful because HTTPS does not automatically mean that a website itself is trustworthy.

A VPN (Virtual Private Network) can provide an additional layer of protection when using public Wi-Fi. A VPN encrypts traffic between the device and the VPN service, which can make it harder for other people on the same Wi-Fi network to monitor the user's network activity. However, users should choose a reputable VPN provider and understand that a VPN does not protect against every type of threat. For example, it does not prevent someone from being tricked by a phishing website or downloading malicious software. (NCSC, Virtual Private Networks, 2021)

It is also important to keep devices up to date. Operating system and application updates often contain security fixes for vulnerabilities that attackers could otherwise exploit. Before using public Wi-Fi, users should make sure their computer, phone or tablet has current security updates. Antivirus or endpoint security software can also provide additional protection against malicious files and programs. (NCSC, Keeping devices and software up to date, 2021)

Another useful precaution is to disable automatic Wi-Fi connections. If a device automatically connects to networks, it has previously used, it may connect without the user realising it. Turning off automatic connection settings gives the user more control over which networks their device joins. Users should also avoid connecting to networks with suspicious names or networks that do not require any authentication when an official network is available. (FTC, Are public wifi networks safe what you need to know, 2023)

File and device sharing should also be disabled when using an unfamiliar network. Features that allow other devices to discover or connect to a computer can increase the amount of information exposed to other people on the same network. Users should check their device's network and sharing settings and use appropriate security settings, particularly when connecting in public places.

6.3 Practical Precautions

Finally, users should remember to log out of important accounts after using a shared or public computer and avoid saving passwords on devices that they do not own. Multi-factor authentication (MFA) should also be enabled on important accounts whenever possible. MFA provides an additional security step, meaning that a stolen password alone may not be enough to access an account. (NCSC, Why MFA matters, 2024b)

In conclusion, public Wi-Fi is useful and convenient, but it should not automatically be considered safe. Users can reduce the risks by connecting only to legitimate networks,

using HTTPS, considering a reputable VPN, keeping devices updated, disabling automatic connections and file sharing, avoiding sensitive activities on untrusted networks, and using strong authentication. By following these practices, people can use public Wi-Fi more safely while reducing the likelihood of their personal information or accounts being compromised.

Overall Conclusion

This report has discussed the various methods of protecting devices from threats as well as the threats in of themselves. The takeaway from this is that the human factor is often the weakest link in cybersecurity as human error is unpredictable and unable to be stopped other than just being vigilant.

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